

KNIFE STEEL COMPARISON

Ten shop staples, side by side. Five dots = best in class. Ratings assume proper heat treatment — a badly treated super-steel loses to a well-treated 1084 every single time.

STEEL	EDGE HOLDING	TOUGHNESS	RUST RESIST	EASE OF SHARPENING	STEEL COST	BEST FOR
1084	● ● ○ ● ●	● ● ● ● ●	● ● ● ● ●	● ● ● ● ●	● ● ● ● ●	First knives, forging classes, hard-use users who maintain their gear
1095	● ● ● ● ●	● ● ● ● ●	● ● ● ● ●	● ● ● ● ●	● ● ● ● ●	Classic carbon EDC + hunters; patina lovers
80CrV2	● ● ● ● ●	● ● ● ● ●	● ● ● ● ●	● ● ● ● ●	● ● ● ● ●	Field knives, camp blades, anything that gets batoned
5160	● ● ● ● ●	● ● ● ● ●	● ● ● ● ●	● ● ● ● ●	● ● ● ● ●	Choppers, swords, big blades that flex and hit
52100	● ● ● ● ●	● ● ● ● ●	● ● ● ● ●	● ● ● ● ●	● ● ● ● ●	Fine-edge hunters + kitchen from a forge; carbon connoisseurs
O1	● ● ● ● ●	● ● ● ● ●	● ● ● ● ●	● ● ● ● ●	● ● ● ● ●	Stock-removal makers with a basic oven; forgiving tool steel
D2	● ● ● ● ●	● ● ● ● ●	● ● ● ● ●	● ● ● ● ●	● ● ● ● ●	Toothy working edges; semi-stainless slicers that stay sharp
AEB-L	● ● ● ● ●	● ● ● ● ●	● ● ● ● ●	● ● ● ● ●	● ● ● ● ●	Kitchen knives, razors — stainless that behaves like carbon
14C28N	● ● ● ● ●	● ● ● ● ●	● ● ● ● ●	● ● ● ● ●	● ● ● ● ●	Budget stainless EDC + kitchen; excellent all-round value
MagnaCut	● ● ● ● ●	● ● ● ● ●	● ● ● ● ●	● ● ● ● ●	● ● ● ● ●	Premium do-everything: hunters, EDC, kitchen, salt water

Carbon vs stainless: under ~11% chromium the steel patinas and rusts — it also usually forges easier and sharpens faster. Pick by the owner's habits, not the spec sheet.

Cost column is bar stock only. Air-hardening + stainless steels also need foil, plates and often cryo — the process cost rides on top of the bar price.

Heat treat decides everything. Ratings assume the recipe on our Heat-Treat Reference sheet, hit within ±25°F. Miss it and every dot on this chart moves.

QUICK PICKS

First knife ever	1084 — cheap, forgiving, teaches everything	Kitchen from the forge	52100 — fine edge, loves a skilled heat treat
Budget stainless that punches up	14C28N — kitchen + EDC, easy money	Kitchen, stainless, refined	AEB-L — thin, hard, screaming sharp
Premium no-excuses steel	MagnaCut — if the customer pays, this is the answer	Toothy work knife	D2 — ugly edge holding that will not quit
Hardest-use fixed blade	80CrV2 or 5160 — toughness over everything	Hamon project	W2 (see Heat-Treat sheet) — shallow hardening on purpose